Siemens SIMATIC S7-1500 System Manual - CPU 1516F-3 PN/DP

Controller Model: 6ES7 516-3FN02-0AB0 Fail-Safe Programmable Logic Controller.

Safety Level: Certified up to SIL 3 according to IEC 62061 and PL e according to ISO 13849-1.

Work Memory: 1.5 MB for program code and 5 MB for process data; 10 ns bit performance.

PROFINET Ports: Integrated 2-port interface X1 (192.168.1.1) and gigabit interface X2 (10.10.20.1).

Primary Function: Cell Master controller managing ROBOT01, CONV01, CAM01, and cell safety interlocks.

Siemens S7-1500 - Fail-Safe Digital Input Addressing & Safety Circuit

F-DI 16x24VDC Module (Slot 5) monitors dual-channel safety devices per schematic E102.

Safety Gate 2 Perimeter Interlock connects to dual-channel safety inputs I:0/10 and I:0/11.

Discrepancy Time: Configured to 250 ms in hardware configuration.

Sensor Evaluation: 1oo2 (1 out of 2) equivalent evaluation with dark test pulses T0 and T1 enabled.

If safety gate opens during automatic motion, safety alarm ALM-205 Safety Gate Open is raised.

Siemens S7-1500 - PROFINET Network Topologies and Diagnostic Buffer

Network Nodes: PLC01 (192.168.1.1), ROBOT01 IRC5 (192.168.1.10), CAM01 Vision (192.168.1.20).

ET 200SP Remote I/O station (192.168.1.30) processes operator console pushbuttons.

Diagnostic Buffer: Read system diagnostic events via TIA Portal online diagnostics or web server.

Cyclic Task Execution: Main OB1 cycle time maintained under 15 ms; Safety OB35 scheduled at 20 ms interval.